

The third point is a matter of practicality. If your tree structure has so many levels of branches that a recursive function would cause a stack overflow before it can reach the leaves, then recursion isn't a suitable solution.

On the other hand, recursion is the best approach for creating programming language compilers. Compiler design is its own expansive subject and beyond the scope of this book. But programming languages have a set of grammar rules that can break source code into a tree structure similar to the way grammar rules can break English sentences into a tree diagram. Recursion is an ideal technique to apply to compilers.

We'll identify many recursive algorithms in this book, and they often have the tree-like structure or backtracking features that lend themselves to recursion well.